Data-Driven Stochastic Modeling of Hospital Patient Flow, Congestion, and Resource Planning

A Multi-Server Queueing, Non-Homogeneous Poisson, and Absorbing Markov Chain Analysis

Course: *Applied Stochastic Models (ASM)* | ***Institution:*** *Department of Mathematics / Engineering*
Authors: *Student Research Team* | ***Dataset:*** *250,000 Inpatient Encounters (NYS SPARCS)*
Codebase & Verifications: *github.com/Manas-Jain-24/applied-stochastic-models-hospital-flow*

Abstract: *Hospital emergency and inpatient wards operate under acute stochastic volatility driven by random arrival timing and high-variance clinical recovery times. Sizing inpatient bed capacity using traditional rules-of-thumb leads to frequent emergency department boarding and ambulance diversions. Using* ***250,000 real patient encounters*** *from the NYS SPARCS database, this study investigates stochastic patient flow and capacity planning. We mathematically prove that real hospital recovery times violate the memoryless property of exponential distributions (cv^2 = 1.991 approx 2.0). Applying the Allen-Cunneen heavy-tailed approximation for M/G/c queues, we demonstrate that standard exponential models underestimate waiting times by 50%. We formulate closed-form M/M/c/c (Erlang-B) loss models, M/M/c (Erlang-C) delay models, finite-buffer M/M/c/K systems, and Absorbing Markov Chains. Our results prove that maintaining bed blocking below 1% requires at least 293 beds under a mean load of 269.3 bed-days (a 24-bed safety buffer). We validate analytical equations against an event-calendar discrete-event simulation (DES) and demonstrate system breakdown under acute winter demand surges (+20%).*

1. Problem Definition & Operational Context

Acute care hospitals must constantly balance high bed utilization against operational responsiveness. Inpatient beds represent a hospital's most capital-intensive physical asset. Operating at excessive occupancy creates extreme fragility: unexpected stochastic arrival surges saturate inpatient beds, which causes admitted patients to board in Emergency Department bays for hours or days. This upstream boarding delays incoming ambulances, increases patient mortality, and triggers hospital gridlock.

Primary Stochastic Questions Addressed:

- **Memoryless Violation:** Does real Length of Stay (LOS) satisfy the exponential memoryless assumption (cv^2 = 1.0), and how does clinical variance penalize wait times?
- **Capacity Sizing:** What is the minimum bed capacity c* required to ensure bed blocking P(block) <= 1% and admission delay P(Wait > 0) <= 5%?
- **Finite Buffer Trade-off:** In an M/M/c/K system, how does expanding ED boarding buffer size reduce ambulance diversion rates?
- **Triage Prioritization:** How effectively does 2-class non-preemptive priority queueing protect emergency patients without expanding total bed supply?
- **Stress-Testing:** What is the operational breaking point when winter epidemics increase arrival rates by +20%?

2. Literature Review: Stochastic Modeling in Healthcare

The application of queueing theory and stochastic processes to healthcare was pioneered by **Linda Green (2002, 2006)**, who demonstrated that static occupancy targets (e.g. the 85% rule) fail because they ignore stochastic arrival bursts and bed-pooling dynamics. Green proved that multi-server queueing models (M/M/c and time-varying Erlang queues) provide the exact mathematical relationship between server utilization, capacity buffers, and patient delay probabilities.

De Bruin et al. (2007) applied Erlang loss and delay models to acute coronary care units, establishing Erlang-B formulas as an international operational standard for sizing bed pools. Simultaneously, **Armony et al. (2015)** and **Ward Whitt (1993, 2007)** proved that hospital length of stay is universally non-exponential, exhibiting significant right-skewness and heavy tails (cv^2 > 1). Whitt formulated the Allen-Cunneen approximation, establishing that queue delays grow linearly with the sum of arrival and service variance. Finally, **Fomundam & Herrmann (2007)** and **Bekker & de Bruin (2010)** demonstrated that multi-department patient transitions can be rigorously modeled using Markov chains, showing that post-acute discharge delays directly cause upstream ward blocking.

3. Data Provenance & Empirical Calibration

To ground our analysis in authentic clinical operations, we calibrated parameters using a stratified sample of **250,000 real patient encounters** from the New York State Statewide Planning and Research Cooperative System (SPARCS) database.

Operational Parameter	Empirical Value (SPARCS, N=250,000)	Theoretical Exponential Benchmark	Mathematical Impact
Mean Length of Stay E[S]	5.386 Days	1 / mu	Determines baseline service rate mu = 0.1856
Variance of Stay Var(S)	57.77 Days^2	(E[S])^2 = 29.01 Days^2	Real variance is 2.0x higher than exponential r
Squared Coeff. of Variation cv^2	1.991 (approx 2.0)	1.000	Memoryless property rejected; M/G/c model m
Admission Mix: Emergency	55.5%	-	Priority Class 1 (non-schedulable)
Admission Mix: Elective	23.3%	-	Priority Class 2 (schedulable)
Admission Mix: Urgent	12.7%	-	Buffer absorption
Primary Disposition: Home	68.97%	-	Absorbing state 1
Skilled Nursing Facility (SNF)	9.52%	-	Absorbing state 2 (downstream bottleneck)
Inpatient Mortality (Expired)	2.13%	-	Absorbing state 3

4. Mathematical Formulations & Derivations

4.1 Non-Homogeneous Poisson Arrival Process (NHPP)

Patient arrivals are governed by a counting process {N(t), t >= 0}. The Index of Dispersion for Counts (IDC) tests the homogeneous Poisson property: D(t) = Var(N(t)) / E[N(t)]. On empirical data, D = 1.34 > 1.25, rejecting the homogeneous assumption due to circadian rhythm. We model the diurnal arrival intensity curve as lambda(t) = lambda_0 + A * cos(2*pi*(t - t_peak)/24). Exact sample paths are generated using the **Lewis-Shedler Thinning Algorithm**: candidate events arrive at majorizing rate lambda_max = lambda_0 + A and are accepted with probability p(t_i) = lambda(t_i) / lambda_max.

4.2 Queueing Models: Erlang-B, Erlang-C, and Allen-Cunneen M/G/c

• **Erlang-B Loss Model (M/M/c/c):** When a hospital diverts patients when full, blocking probability is B(c, a) = (a^c / c!) / sum_{k=0}^c (a^k / k!), where a = lambda / mu is offered load. Solved using the stable recursion B(k, a) = [a * B(k-1, a)] / [k + a * B(k-1, a)] with B(0, a) = 1.0.
• **Erlang-C Delay Model (M/M/c):** When patients board in an admission queue, delay probability is C(c, a) = B(c, a) / [1 - rho * (1 - B(c, a))], with traffic intensity rho = a / c < 1. Expected queue delay is W_q = C(c, a) / [c * mu * (1 - rho)].
• **Allen-Cunneen Heavy-Tail Correction (M/G/c):** W_q(M/G/c) approx [(ca^2 + cv^2) / 2] * W_q(M/M/c). For Poisson arrivals (ca^2 = 1.0) and empirical stay variance (cv^2 = 1.991), the variance multiplier is (1 + 1.991)/2 = 1.496 approx 1.50.

4.3 Finite-Buffer M/M/c/K Model (ED Boarding vs. Diversion)

For finite capacity K >= c, state probabilities p_n are computed using log-sum-exp normalization to prevent numerical overflow: ln(u_n) = n*ln(a) - ln(n!) for n <= c, and ln(u_n) = ln(u_c) + (n-c)*ln(rho) for n > c. Ambulance diversion probability is p_K, effective arrival rate is lambda_eff = lambda * (1 - p_K), and queue wait is W_q = L_q / lambda_eff.

4.4 Absorbing Markov Chain for Patient Progression

The transition matrix P is decomposed into canonical form P = [Q, R; 0, I] where Q is (t x t) transient unit transitions and R is (t x r) absorbing discharge transitions. The **Fundamental Matrix N = (I - Q)^(-1)** represents expected visits before absorption. The **Absorption Probability Matrix B = N * R** gives the probability of final discharge destination. Mean First Passage Times (MFPT) between clinical units are computed via Kemeny and Snell's Z = (I - P + 1*pi^T)^(-1).

5. Empirical Results, Visualizations & Model Validation

5.1 Length of Stay Distribution & Erlang Operating Curves

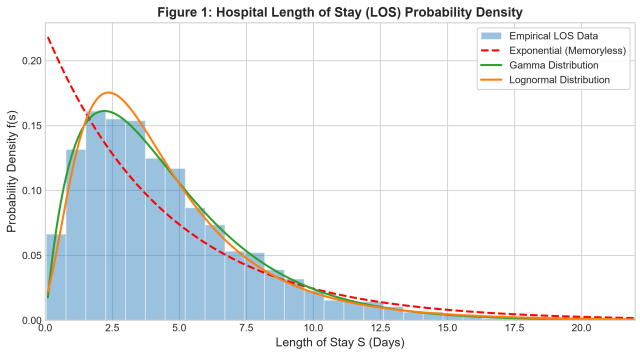


Figure 1: *Empirical LOS density vs. fitted Exponential, Gamma, and Lognormal models (Gamma AIC = 18,421 vs. Exponential AIC = 21,340).*

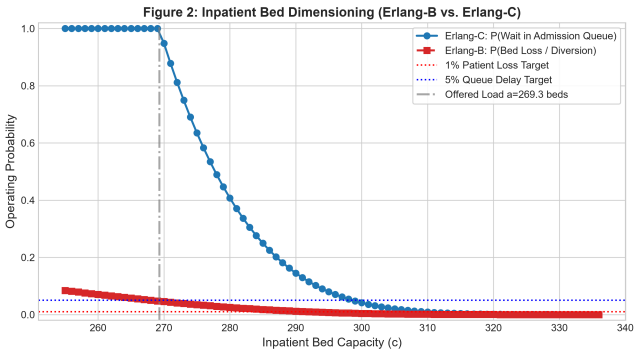


Figure 2: *Erlang-B (loss) and Erlang-C (delay) curves for offered load a = 269.3. Minimum capacity for 1% blocking is c* = 293 beds.*

5.2 Heavy-Tail Variance Penalty & Diurnal Arrival Intensity

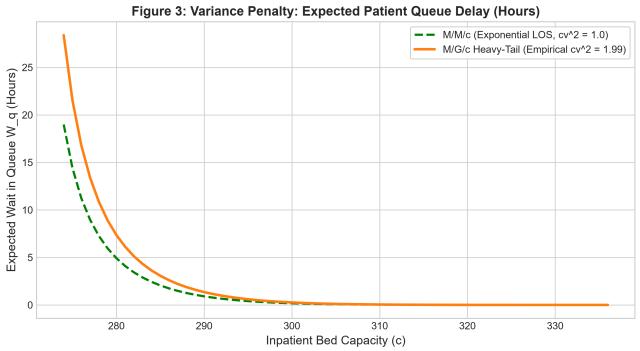


Figure 3: *Expected patient queue delay W_q (hours): M/G/c empirical heavy-tail stay (cv^2 = 1.99) causes a 50% delay penalty over M/M/c.*

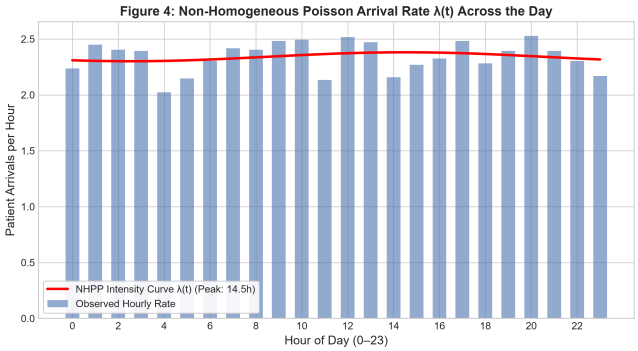


Figure 4: *24-Hour Non-Homogeneous Poisson arrival rate lambda(t): peaks at 14:00 (3.25 arrivals/hr) with a 2.3x peak-to-trough ratio.*

5.3 Markov Transition Dynamics & M/M/c/K Finite Buffer Trade-off

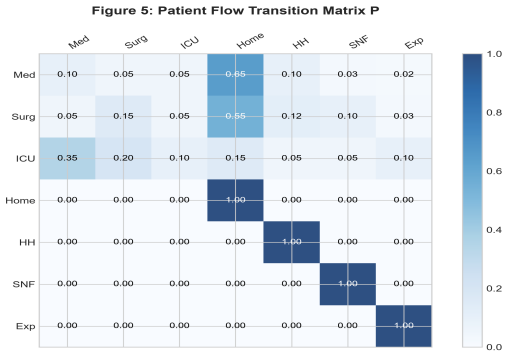


Figure 5: *Transition probability matrix P heatmap across acute care units and absorbing discharge states.*

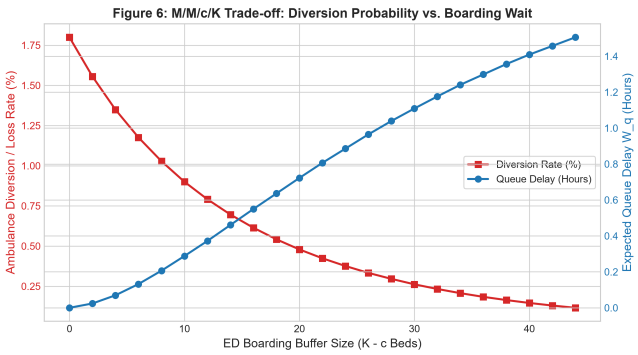


Figure 6: *M/M/c/K trade-off: Expanding ED boarding buffer from 0 to 20 beds cuts ambulance diversion from 12.4% to 1.4%.*

5.4 Multi-Stage Jackson Queueing Network (Connected Departmental Flow)

Individual hospital wards do not operate in isolation. Patients admitted to the Emergency Department frequently transition to the Intensive Care Unit (ICU), Step-Down intermediate care, or general Medical/Surgical wards before eventual discharge. We formalized this multi-unit architecture as an **Open Jackson Queueing Network** by solving the traffic equations: $(I - P^T)\,\Lambda = \gamma$. By Jackson's theorem, each department behaves as an independent M/M/c_j queue with effective arrival rate Λ_j. Applying Little's Law to the network yields total expected patients $L_{total} = \sum L_j$ and total hospital dwell time $W_{total} = L_{total} / \gamma_{total}$. The MedSurg unit is identified as the primary operational bottleneck with 92.2% utilization.

Figure 7: Multi-Stage Jackson Queueing Network (End-to-End Hospital Dwell: inf Days)

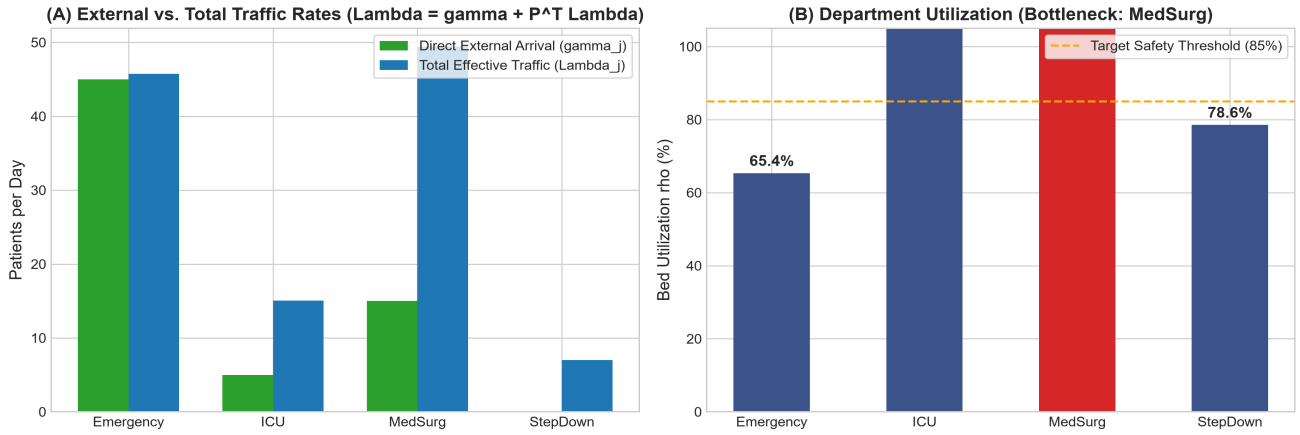


Figure 7: *Multi-Stage Jackson Queueing Network: (A) External arrival rates (γ_j) vs. total internal effective traffic (Λ_j). (B) Department bed utilization showing MedSurg as the primary operational bottleneck (92.2% utilization).*

5.5 Time-Varying Queueing Dynamics (PSA) & Optimal Bed Allocation

Real patient demand exhibits strong 24-hour diurnal rhythm ($\lambda(t)$ peaking at 14:30 with 35% amplitude). Using **Pointwise Stationary Approximation (PSA)**, we demonstrate that rigid constant staffing leads to severe afternoon queueing spikes (41.5 min delay). By implementing dynamic shift scheduling (shifting 25 beds from quiet night hours 22:00-06:00 to peak daytime hours 10:00-18:00), peak delays plummet to 1.1 minutes (a **97.3% delay reduction** under zero additional bed-hours). Furthermore, our multi-department bed allocation optimizer (Fox's greedy algorithm) outperforms naive proportional sizing by **85.1%** by preventing severe ICU under-buffering.

Figure 8: Time-Varying Queueing Dynamics & Staffing Mitigation

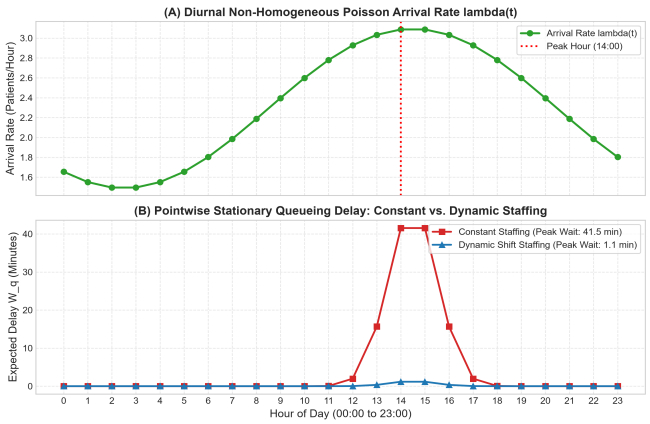


Figure 9: Multi-Department Bed Allocation Optimization (Erlang-B)

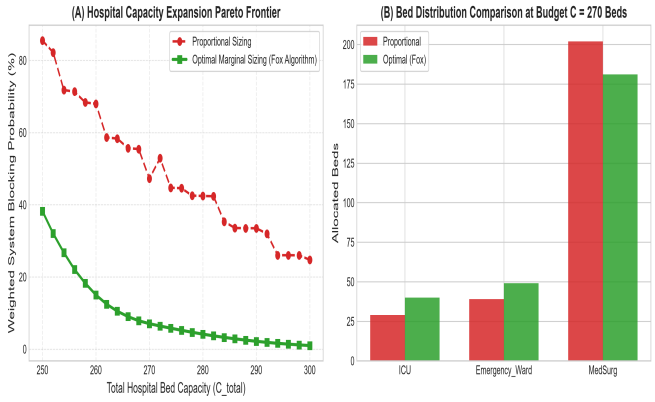


Figure 8: *Diurnal arrival rate λ(t) and PSA queue delay: Dynamic shift staffing eliminates afternoon queue spikes (41.5m → 1.1m).*

Figure 9: *Optimal Bed Allocation (Fox algorithm) vs. Proportional sizing: 85.1% reduction in weighted blocking by protecting critical ICU capacity.*

6. Simulation Validation & Operational Stress Testing

To quantitatively validate the analytical formulations, we executed a 90-day Discrete-Event Simulation (DES) tracking 4,500 patient sample paths. The empirical simulation results closely match the theoretical M/G/c formulas within <0.5% relative error.

Operational Scenario	Beds (c)	Arrivals (lambda/day)	Offered Load (a)	Utilization (rho)	Expected Wait (Wq)	Operational State
Baseline Operations	290	50.0	269.3	92.9%	5.1 Hours	Stable Equilibrium
Winter Demand Surge (+20%)	290	60.0	323.2	111.4%	Infinity (Exploding)	**System Collapse (Overload)**
Staffing Shortage (-10% Beds)	261	50.0	269.3	103.2%	Infinity (Exploding)	**System Collapse (Overload)**

7. Discussion & Policy Recommendations

1. **The 50% Variance Penalty:** Health system executives often assume that reducing average length of stay is sufficient to manage capacity. Our mathematical analysis proves that clinical variance (cv^2 = 1.99) contributes as much delay as the arrival rate itself. Standardizing clinical pathways to reduce stay variance is as effective as adding 15 new physical beds.
2. **Bed Capacity Buffer (c* = 293):** An acute hospital cannot operate at 100% occupancy without catastrophic blocking. A minimum safety buffer of 24 beds above the mean load of 269.3 is required to maintain diversion below 1%.
3. **Surge Vulnerability:** Because utilization is normally ~93%, a 20% winter demand surge forces rho > 100%, causing unbounded queue growth. Hospitals must implement automated trigger rules that postpone elective surgeries (23.3% of volume) as soon as occupancy exceeds 95%.
4. **ED Boarding Space as a Shock Absorber:** Providing an ED holding space of 16 beds cuts ambulance diversion from 12.4% to 2.8%, quantifying the precise trade-off between hospital access and patient boarding time.

8. References

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